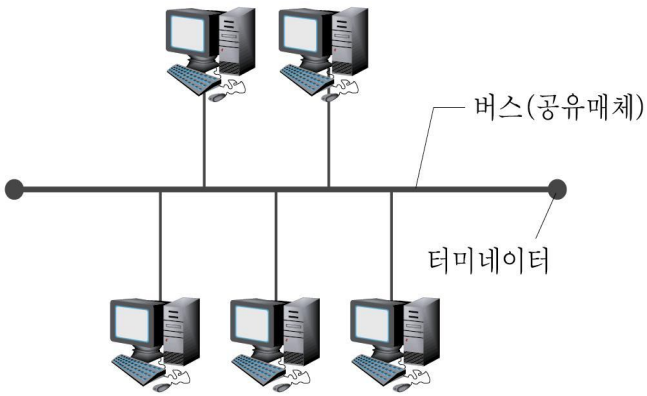
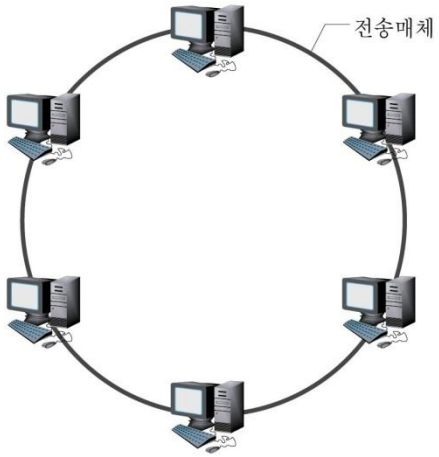
LAN

	Ethernet/802.3 LAN	TokenRing/802.5 LAN
기관	1980년 DEC, Intel, Xerox 표준발표 IEEE에 의해 표준으로 정의	1972년 IBM사에서 처음 고안 IEEE에 의해 표준으로 정의
토폴로지	버스형, 스타형	링형
기본 전송매체	10 BASE 2 10 BASE 5 10 Base T	10 BASE 2 10 BASE 5
매체접근 방식	CSMA/CD	Token Passing

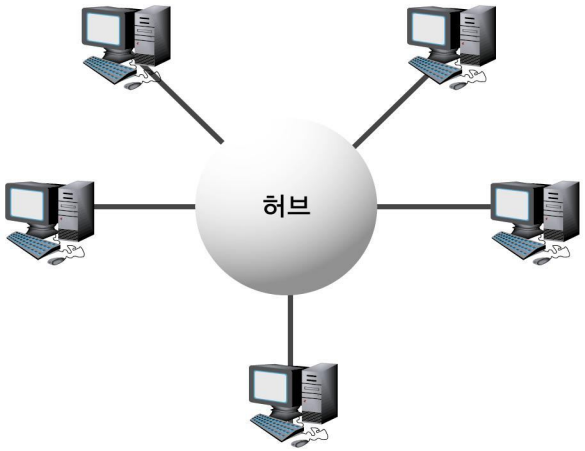
Network topology



Ethernet



Token Ring, FDDI



Fast Ethernet

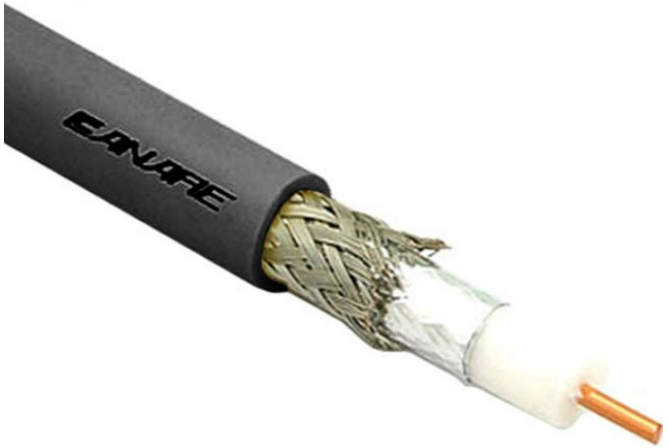
Gigabit Ethernet

100 BASE FX

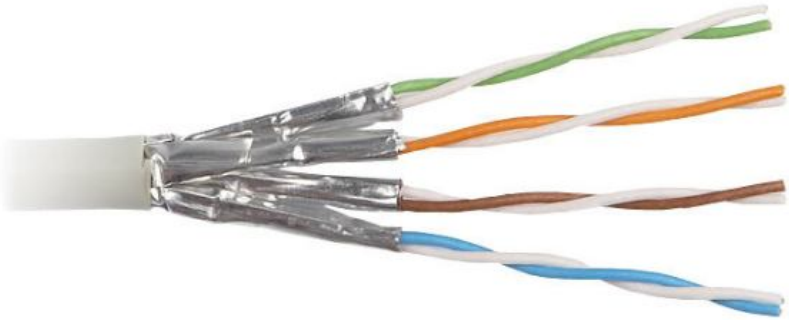
속도 채널 케이블 타입

	10Base5	10BaseT	100BaseTX	100BaseFX
Mex. Segment Length	500 meters	100 meters	100 meters	400 meters
Topology	Bus	Star	Star	Point-to-Point

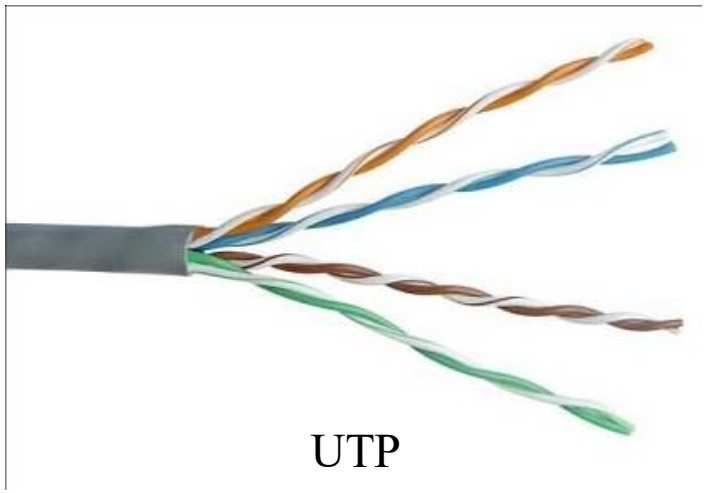
10 Base 2



10 Base 5



STP

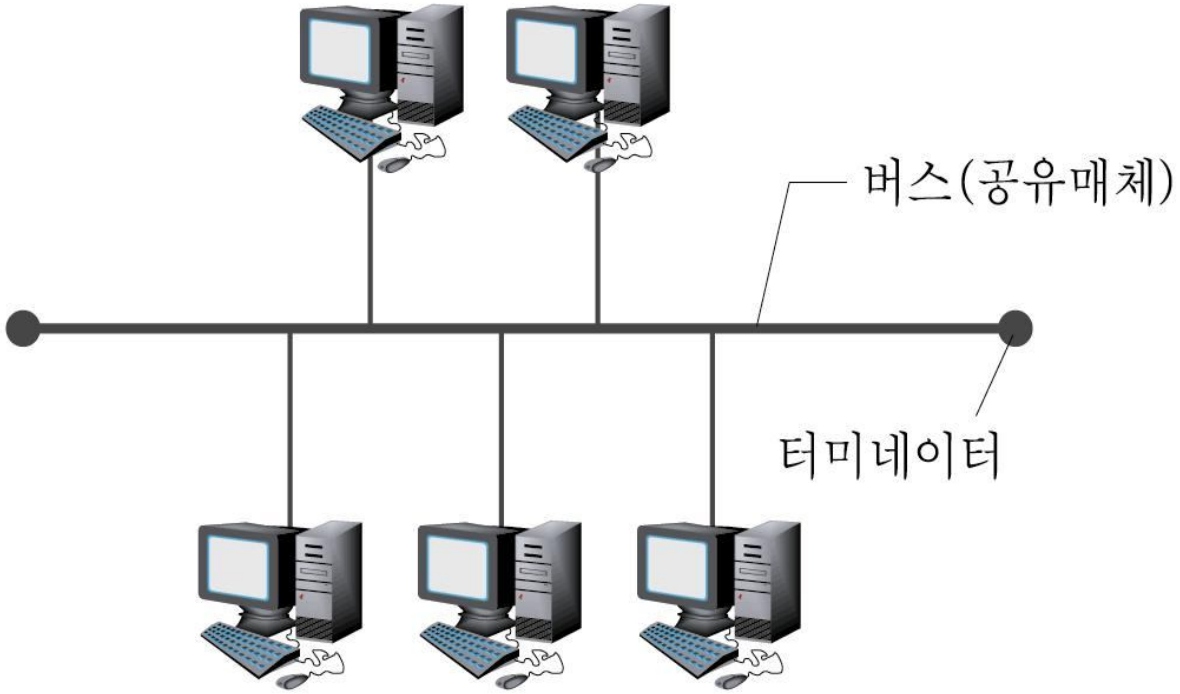


UTP

10 Base T

Bus Topology

- 가장 간단하며 단일 케이블에 의해 네트워크의 모든 컴퓨터 연결
- 모든 노드들은 Bus에 T자형으로 연결되며 버스 끝은 터미네이터를 달아 신호의 반사를 방지



유선 전송 매체

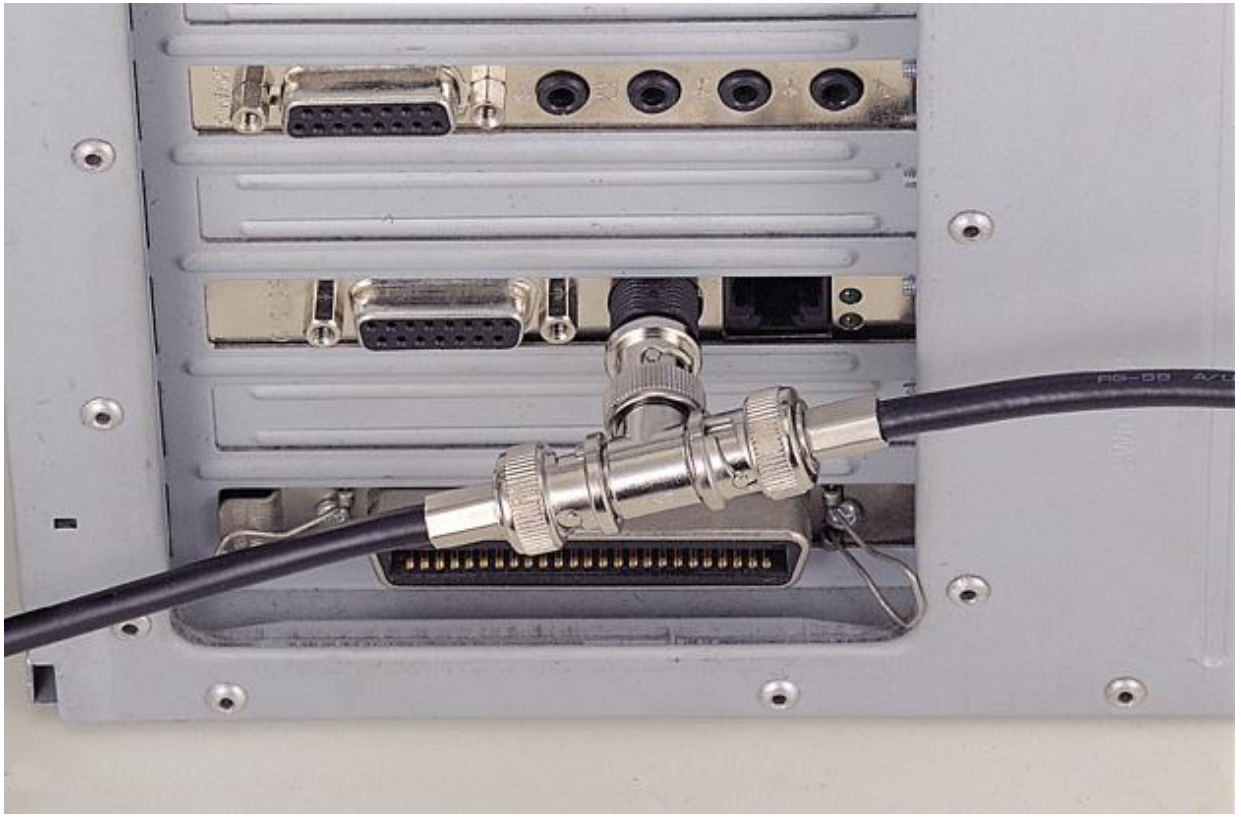
랜 케이블 - BNC

Barrel T Terminator

BNC 커넥터의 종류

유선 전송 매체

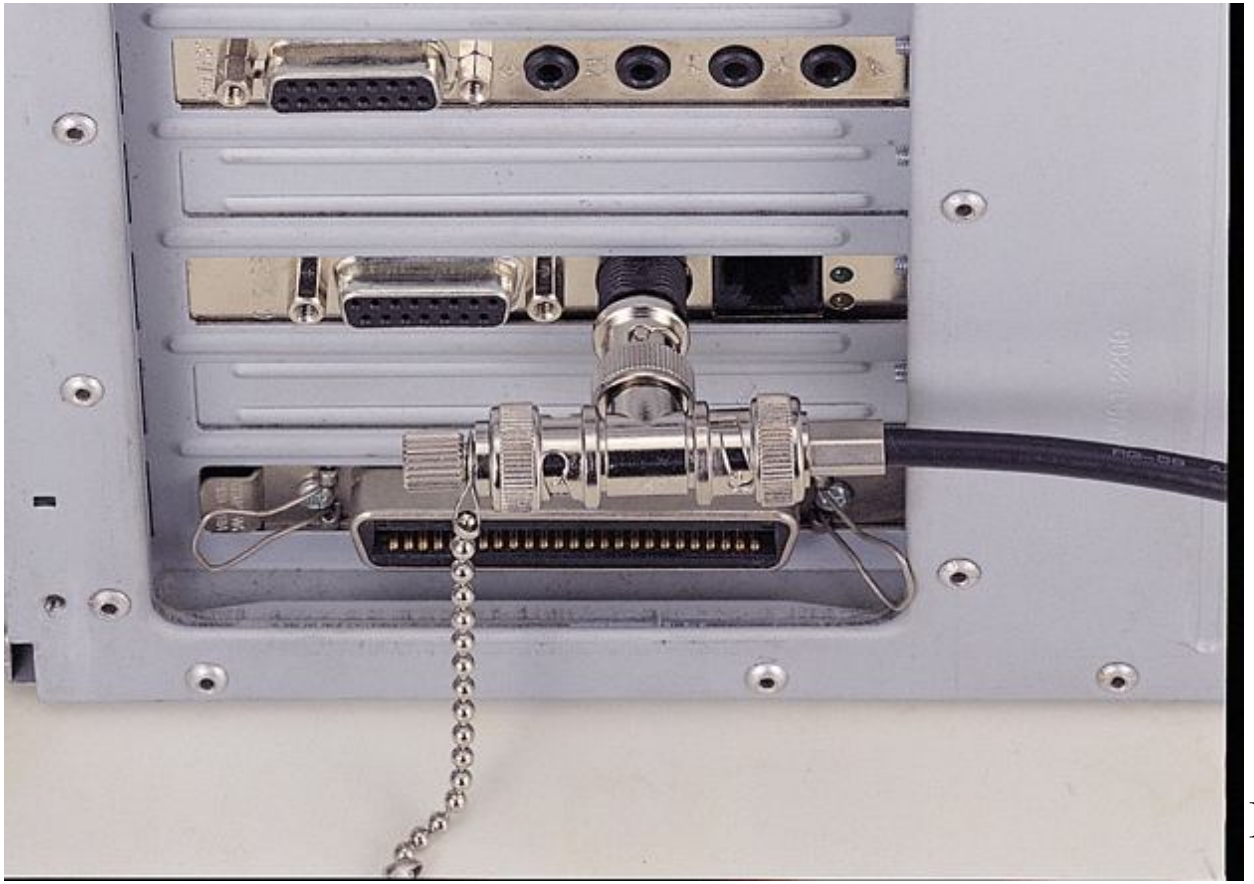
랜 케이블 - BNC



BNC
T-커넥터연결

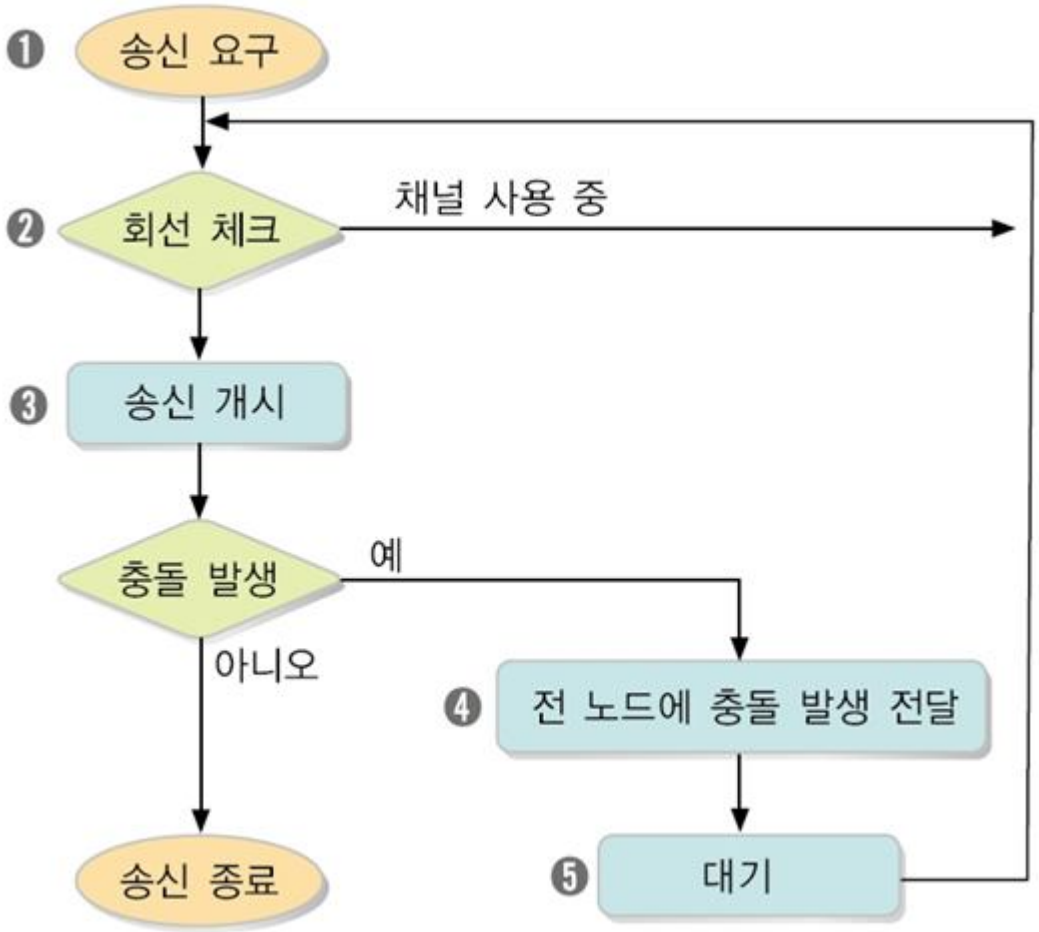
유선 전송 매체

랜 케이블 - BNC

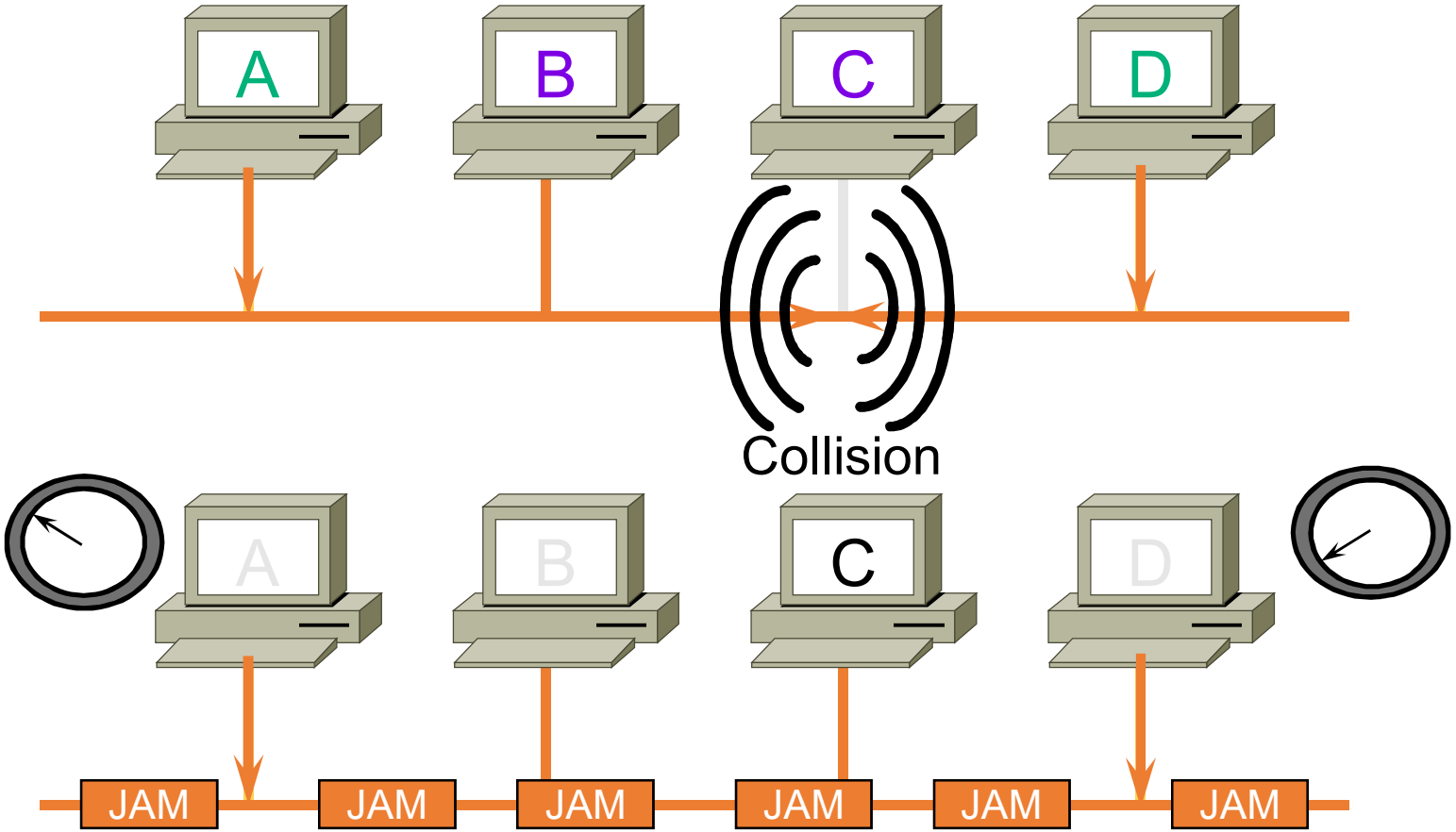


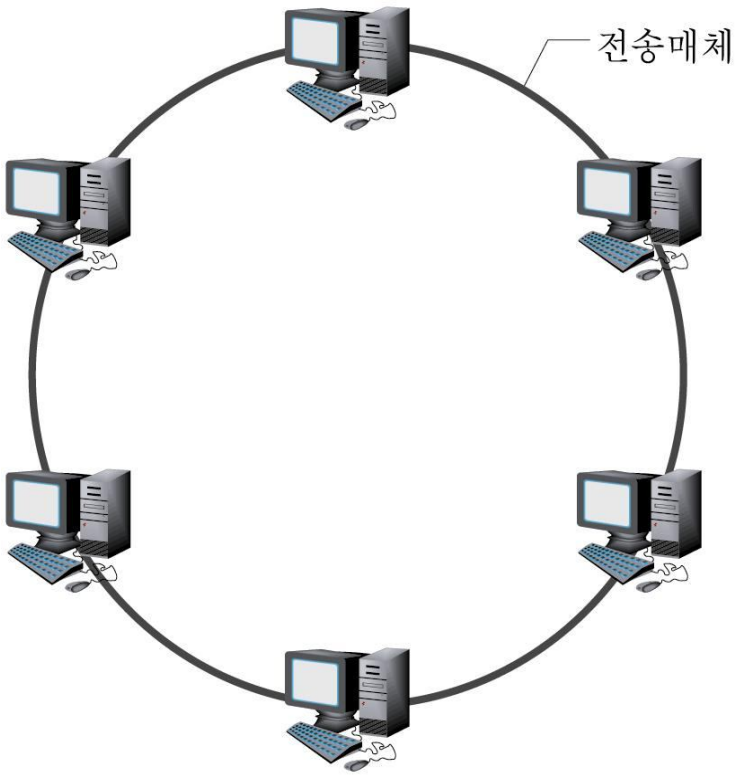
BNC Terminator

CSMA/CD



CSMA/CD



Token Passing

- 채널을 전용하는 자격증인 토큰을 미리 정해진 순서에 따라 주고받으면서 전송하는 방법
- Free Token 과 Busy Token 사용